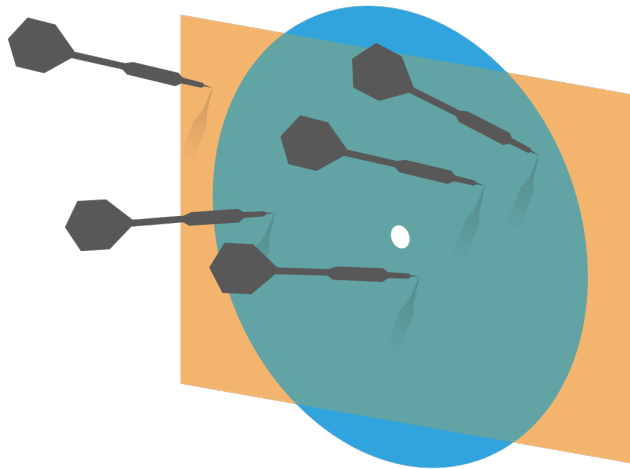


Probability and Statistics for Computer Science



“Probabilistic analysis is mathematical, but intuition dominates and guides the math” – Prof. Dimitri Bertsekas

Credit: wikipedia

Last time

- ✱ IQR, Mode

- ✱ Scatter plots for relationships

- ✱ Correlation Coefficient

$$\hat{y}^p = r \cdot \hat{x}$$

Prediction Formula

✱ In standard coordinates

$$\hat{y}_0^p = r \hat{x}_0 \quad \text{where} \quad r = \text{corr}(\{(x_i, y_i)\})$$

✱ In original coordinates

$$\frac{y_0^p - \text{mean}(\{y_i\})}{\text{std}(\{y_i\})} = r \frac{x_0 - \text{mean}(\{x_i\})}{\text{std}(\{x_i\})}$$

Root-mean-square (RMS) prediction error



Given $var(\{u_i\}) = 1 - 2ar + a^2$
& $a = r$

$$var(\{u_i\}) = 1 - r^2$$



$$\begin{aligned} RMS \text{ error} &= \sqrt{mean(\{u_i^2\})} \\ &= \sqrt{var(\{u_i\})} \\ &= \sqrt{1 - r^2} \end{aligned}$$

Objectives

- ✱ Probability a first look
 - ✱ Outcome and Sample Space
 - ✱ Event
 - ✱ Probability: Axioms & Properties
 - ✱ Calculating probability *using counting*

Warm up



Warm up : Counting

Addition Principle

Multiplication Principle

Example

① Ans: $4 + 4 + 4 = 12$

addition rule

② Ans: $4 \times 4 \times 4 = 64$

multiplication rule

draw one card
out of 52-set

① 3 or 6 or 1

② $3 \times 6 \times 1$

draw three with
replacement.

Warm up (1)

10 different gifts to 10 friends,
each gets 1 gift.

of arrangement = ?

$$10 \times 9 \times \dots \times 1 = 10!$$

Warm up (11)

4 different gifts to 10 friends,
each gets 1 gift.

th of arrangement = ?

$$10 \times 9 \times 8 \times 7 = \frac{10!}{6!}$$

$$\frac{10 \times 9 \times 8 \times 7 \times (6 \times 5 \cdots \times 1)}{6 \times 5 \cdots \times 1}$$

Warm up (III)

4 identical gifts to 10 friends,
each gets 1 gift.

of arrangement = ?

$$\frac{10!}{6! \textcircled{4!}}$$

$$\binom{10}{4}$$

Commutative

$$A \cap B = B \cap A$$

$$A \cup B = B \cup A$$

Associative

$$(A \cap B) \cap C = A \cap (B \cap C)$$

$$(A \cup B) \cup C = A \cup (B \cup C)$$

Distributive

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Idempotent

$$A \cap A = A$$

$$A \cup A = A$$

Identity

$$A \cup \emptyset = A$$

$$A \cap \emptyset = \emptyset$$

$$A \cup \Omega = \Omega$$

$$A \cap \Omega = A$$

Complement

$$A \cup A^c = \Omega$$

$$A \cap A^c = \emptyset$$

$$\Omega^c = \emptyset$$

$$\emptyset^c = \Omega$$

De Morgan's

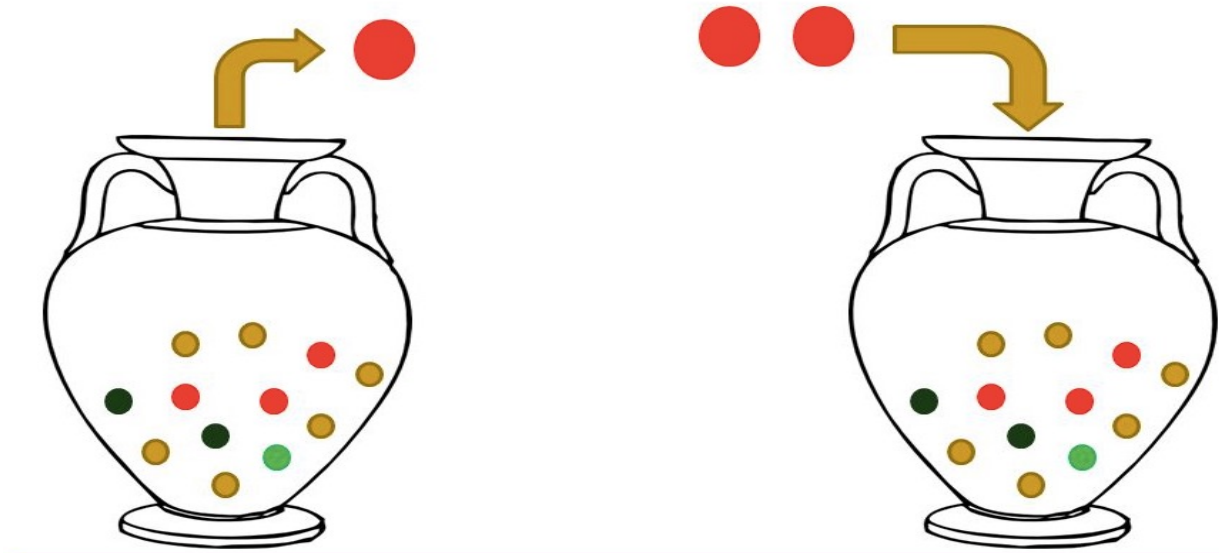
$$(A \cap B)^c = A^c \cup B^c$$

$$(A \cup B)^c = A^c \cap B^c$$

Outcome

- ✱ An outcome **A** is a possible result of a random repeatable experiment

Random:
uncertain,
Nondeterministic, ...



Sample space

✱ The Sample Space, Ω , is the set of all possible mutually exclusive outcomes associated with the experiment

disjoint

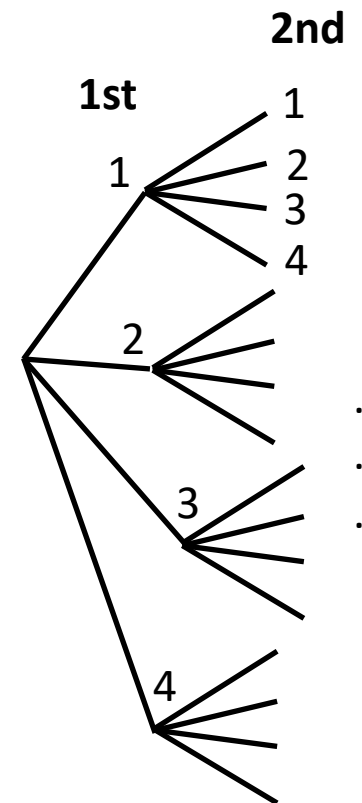
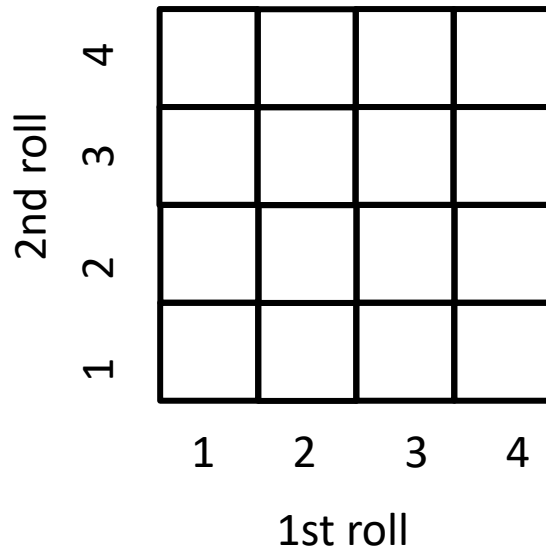
✱ Discrete or Continuous

Sample Space example (1)

✱ Experiment: we roll a 4sided-die twice

✱ **Discrete** Sample space:

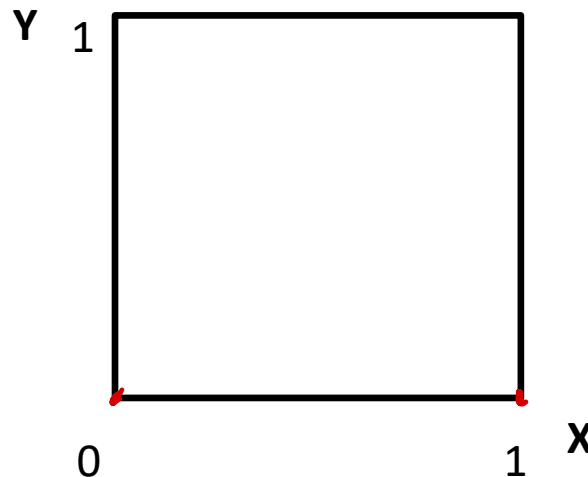
$\{(1,1), (1,2), \dots\}$



Sample Space example (2)

- ✱ Experiment: Romeo and Juliet's date
- ✱ **Continuous** Sample space:

$$\Omega = \{ (x, y) \mid 0 \leq x, y \leq 1 \}$$



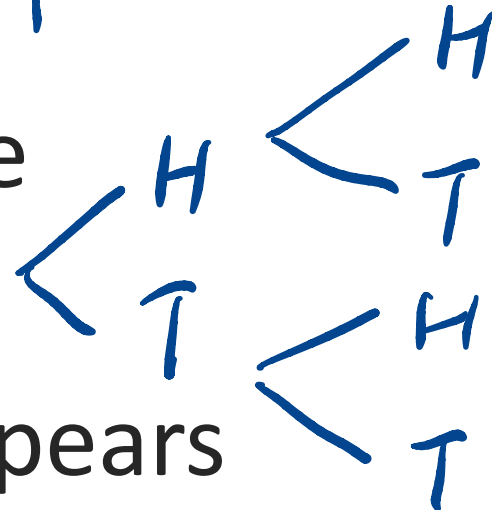
Sample Space depends on experiment (3)

* Different coin tosses

* Toss a fair coin



* Toss a fair coin twice



* Toss until a head appears



Sample Space depends on experiment (4)

- ✱ Drawing 2 socks one at a time from a bag containing 1 blue sock, 1 orange sock and 1 white sock **with replacement**?

$$3 \times 3 = 9 \quad \{ B, O, W, B, O, W, B, O, W \}$$

- ✱ Drawing 2 socks one at a time from a bag containing 1 blue sock, 1 orange sock and 1 white sock **without replacement**?

$$3 \times 2 = 6$$

Q1. Sample Space

✱ Drawing 2 socks one at a time from a bag containing 1 blue sock, 1 orange sock and 1 white sock **with replacement**? What is the number of unique outcomes in the sample space?

A. 5 B. 7 C. 9

Q2. Sample Space

✱ Drawing 2 socks one at a time from a bag containing 1 blue sock, 1 orange sock and 1 white sock **without replacement**? What is the number of unique outcomes in the sample space?

- A. 5 **B. 6** C. 9

Sample Space in real life



- ✱ Possible outages of a power network
- ✱ Possible mutations in a gene
- ✱ A bus' arriving time
- ✱ Possible 3-4 pl teams in CS361

Event

- * An event $\mathbf{E}$ is a subset of the sample space Ω
- * So, an event is a set of outcomes that is a subset of Ω , ie.
 - * Zero outcome
 - * One outcome
 - * Several outcomes
 - * All outcomes

The same experiment may have different events

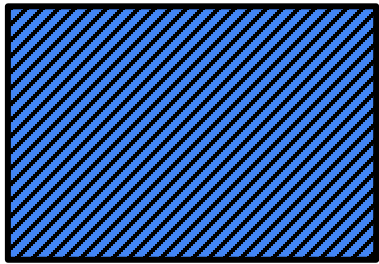
- ✱ When two coins are tossed $E: \{HH, TT\}$
- ✱ Both coins come up the same?
- ✱ At least one head comes up?

$$E: \{HH, HT, TH\}$$

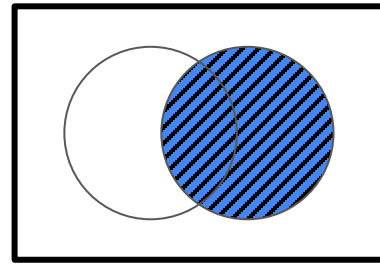
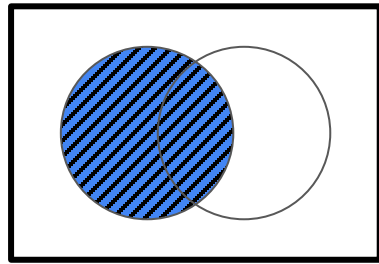
Some experiment may never end

- ✱ Experiment: Tossing a coin until a head appears
- ✱ **E:** Coin is tossed at least 3 times
This event includes infinite # of outcomes

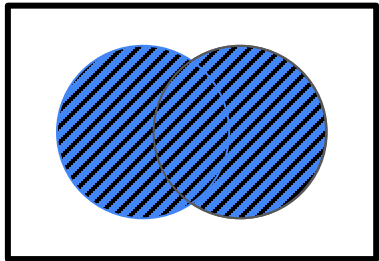
Venn Diagrams of events as sets



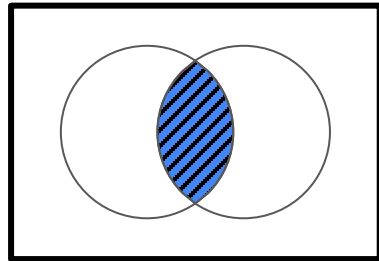
$\{\Omega \cup \{>1\}\} E_1$



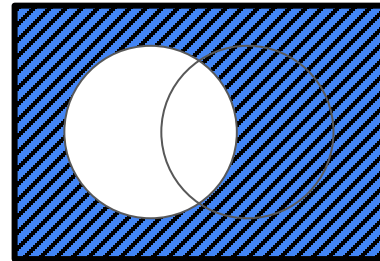
E_2



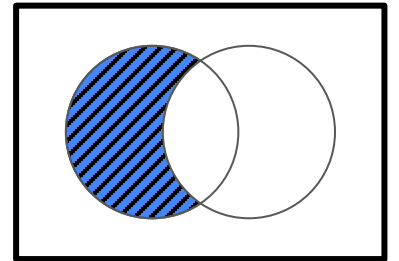
$E_1 \cup E_2$



$E_1 \cap E_2$



E_1^c



$E_1 - E_2$

Combining events

✱ Say we roll a six-sided die. Let

$$E_1 = \{1, 2, 5\} \text{ and } E_2 = \{2, 4, 6\}$$

Do this at home

✱ What is $E_1 \cup E_2$

✱ What is $E_1 \cap E_2$

✱ What is $E_1 - E_2$

✱ What is $E_1^c = \Omega - E_1$

Frequency Interpretation of Probability

- ✱ Given an experiment with an outcome **A**, we can calculate the probability of **A** by repeating the experiment over and over

$$P(A) = \lim_{N \rightarrow \infty} \frac{\text{number of time } A \text{ occurs}}{N}$$

- ✱ So,

$$0 \leq P(A) \leq 1$$
$$\sum_{A_i \in \Omega} P(A_i) = 1$$

Axiomatic Definition of Probability

✱ A probability function is any function **P that maps sets to real number** and satisfies the following **three** axioms:

1) Probability of any event E is non-negative

$$P(E) \geq 0$$

2) Every experiment has an outcome and

$$P(\Omega) = 1$$

Axiomatic Definition of Probability

3) The probability of the union of disjoint events is additive

$$P(E_1 \cup E_2 \cup \dots \cup E_N) = \sum_{i=1}^N P(E_i)$$

$\cup \{E_1, E_2, E_3, \dots\}$

if $E_i \cap E_j = \emptyset$ for all $i \neq j$

i.e. $\downarrow$ $P(\{2\} \& \{4\}) = 0$

$$E_1 = \{2\}$$

$$E_2 = \{4\}$$

$$P(E_1 \cap E_2) = 0$$

Q3. Disjoint/Mutual Exclusive

✱ Toss a coin 3 times

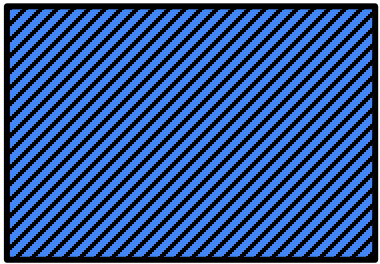
The event “exactly 2 heads appear” and “exactly 2 tails appear” are disjoint.

☒ A. True

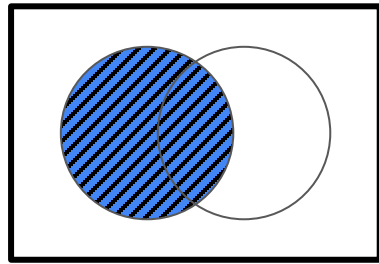
B. False

~~$\{HH TT\}$~~

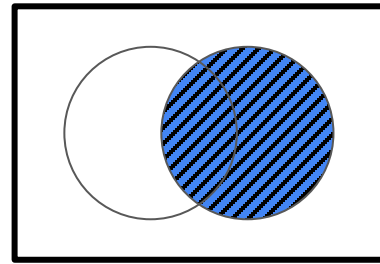
Venn Diagrams of events as sets



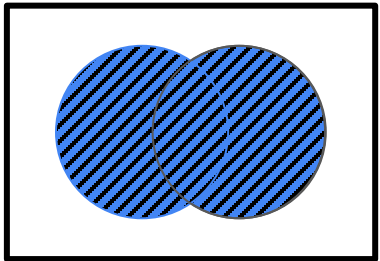
Ω



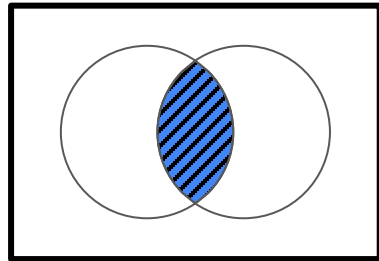
E_1



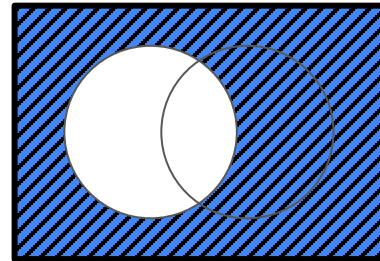
E_2



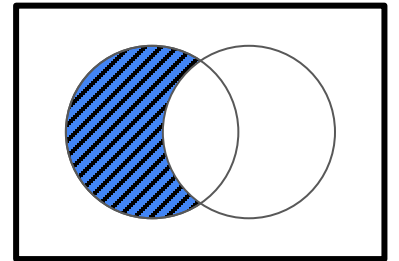
$E_1 \cup E_2$



$E_1 \cap E_2$



E_1^c

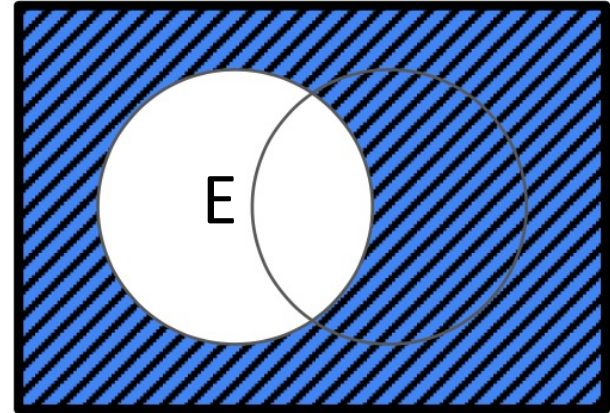


$E_1 - E_2$

Properties of probability

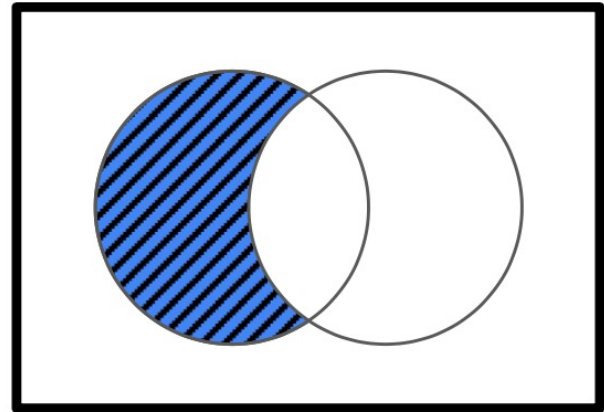
✱ The complement

$$P(E^c) = 1 - P(E)$$



✱ The difference

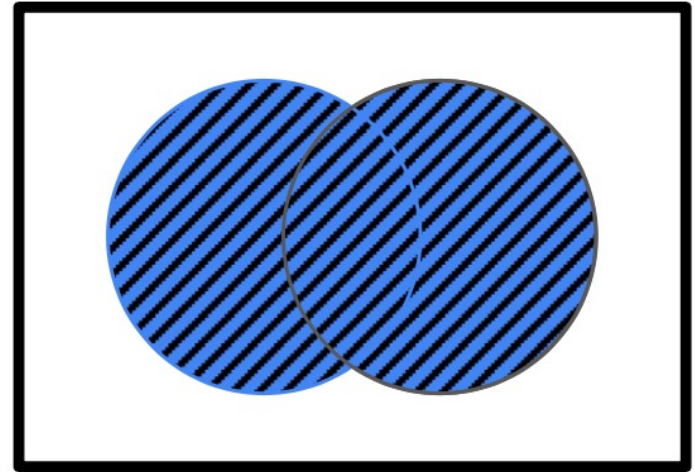
$$P(E_1 - E_2) = P(E_1) - P(E_1 \cap E_2)$$



Properties of probability

✱ The union

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$



✱ The union of multiple E

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_2 \cap E_3) - P(E_3 \cap E_1) + P(E_1 \cap E_2 \cap E_3)$$

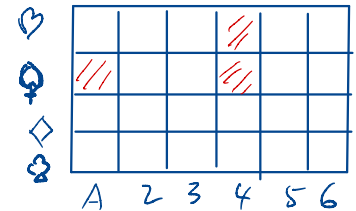
The Calculation of Probability

- ✱ Discrete countable finite event
- ✱ Discrete countable infinite event
- ✱ Continuous event

Counting to determine probability of countable finite event

- ✱ From the last axiom, the probability of event **E** is the sum of probabilities of the disjoint outcomes

$$P(E) = \sum_{A_i \in E}^{N_E = |E|} P(A_i)$$



- ✱ If the outcomes have equal probability,

$$P(E) = \frac{\text{number of outcomes in } E}{\text{total number of outcomes in } \Omega} = \frac{|E|}{|\Omega|}$$

Probability using counting: (1)

✱ Tossing a fair coin twice:

✱ Prob. that it appears the same?

$$E : \{HH, TT\}$$

$$|E| = 2$$

$$\Omega : \{ \quad - \quad - \quad - \quad \}$$

$$|\Omega| = 4$$

✱ Prob. that at least one head appears?

Probability using counting: (2)

✱ 4 rolls of a 5-sided die:

E: they all give different numbers

✱ Number of outcomes in the sample space

$$5 \times 5 \times 5 \times 5$$

$$|\Omega| = 5^4$$

✱ Number of outcomes that make the event happen:

$$|E| = 5 \times 4 \times 3 \times 2$$

✱ Probability:

$$P(E) = \frac{5!}{5^4}$$

Probability using counting: (2)

✱ What about $N-1$ rolls of a N -sided die?

E: they all give different numbers

✱ Number of outcomes in the sample space

✱ Number of outcomes that make the event happen:

✱ Probability:

Additional References

- ✱ Charles M. Grinstead and J. Laurie Snell
"Introduction to Probability"
- ✱ Morris H. Degroot and Mark J. Schervish
"Probability and Statistics"

Assignments

- ✱ Work on 2nd half of Module Week2
- ✱ Quiz1 and Quiz2 due Sat. 11:59pm
- ✱ HW1 due 11:59PM Thurs. night (tonight)

See you next time

*See
You!*

